

Critical Complements in Fabius Conditioning

Sharp information loss, geometric phase constants,
and a logarithmic crossover

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Abstract

Let $S_q = \sum_{j \geq 1} q^j U_j$, with independent uniform coordinates and $0 < q < 1$. Condition the original law on S_q lying below its mean under a large exponential tilt. The repository's earlier conditioning papers identify when a coordinate block resembles its tilted product law, but leave the critical-complement regime without sharp rates. We resolve its geometric-prefix version: observe $n - m$ bulk coordinates, where n is the effective dimension and $m = o(n)$.

For fixed m , the overlap with the tilted product is $[\frac{1}{2} \log(2\pi n) + 1 + K_{q,\rho,m}] / \sqrt{2\pi n} + o(n^{-1/2})$, with an explicit geometric-phase constant. Relative entropy is $\frac{1}{2} \log(2\pi n) - h(R_m + E) + O(n^{-1})$, and we calculate its $1/n$ coefficient. An exact compact boundary correction explains the slow convergence of the overlap expansion.

For $m \rightarrow \infty$, $m = o(n)$, relative entropy becomes $\frac{1}{2} \log(n/m) - \frac{1}{2} + o(1)$. Total variation has a different, three-regime structure. Its overlap crosses from $\log n / \sqrt{n}$ to $\sqrt{(m/n) \log(n/m)}$ at $m \asymp \log n$. We give one leading formula throughout, using the two real branches of Lambert W and the Gamma rate function. The information discarded by a fixed complement is exactly an exponential-noise mutual information in the limit.

Proofs use an exact one-dimensional likelihood identity and a signed Gamma-convolution calculus, not an unquantified Gaussian extrapolation. The package includes reproducible floating-point diagnostics and a further-research program.

Status. Conventional proofs in a research draft; not Lean-certified or externally refereed. The resolved question is a specified repository frontier, not a claimed solution of a named famous conjecture. Classical Gibbs conditioning, log-concavity, and Gamma asymptotics are not claimed as new. Worldwide priority has not been established.

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1 The question and its place in the repository

1.1 From a qualitative endpoint to sharp asymptotics

The probabilistic Fabius model is unusually well suited to studying rare configurations. The event that a geometrically weighted sum of independent uniform variables is very small constrains many coordinates simultaneously. An exponential tilt removes most of the rarity, but conditioning on the sum still couples the coordinates. The question is how much of that coupling an observer can detect.

Two research manuscripts in the inspected ProveIt snapshot already address this problem. *Sharp Conditioning Laws for Uniform Random Series* proves a variance-fraction criterion for general positive summable weights, computes total-variation and entropy limits when the observed variance fraction tends to a number strictly between zero and one, and proves only the qualitative conclusions $d_{TV} \rightarrow 1$, $D_{KL} \rightarrow \infty$ when that fraction tends to one. Its further-research list and conclusion explicitly leave critical complements and quantitative corrections open [1]. A companion lacunary-series manuscript develops full-configuration approximations, marginal corrections, fluctuation processes, and boundary laws [2].

The present paper asks the following precise continuation question:

For geometric weights, how fast do the marginal total variation and relative entropy approach their singular endpoints when the unobserved bulk contains only $m = o(n)$ coordinates? What survives of the geometric phase when m stays bounded?

The answer is not obtained by merely inserting a moving variance fraction into a fixed-fraction limit. Relative entropy has a universal diverging logarithm once $m \rightarrow \infty$, whereas overlap has a transition at $m \asymp \log n$. Fixed complements retain a non-Gaussian boundary density and explicit phase constants.

Our inspection is pinned to 0a543d5e435df885f78ca731d71856f3ee0ab2d4. The relevant predecessor paths and precise comparison are recorded in the bibliography and the package's `provenance.json`. The repository is large; this is a targeted comparison of the pertinent conditioning manuscripts, not an exhaustive verification of every research report. The proofs below do not require assuming the correctness of the unformalized predecessor theorems.

1.2 Prior work and the contribution boundary

Conditional approximation by exponential families is classical. Diaconis and Freedman studied conditional limit theorems and finite versions of de Finetti's theorem [3]. Dembo and Zeitouni developed refinements of the Gibbs conditioning principle, including growing blocks [4]. Those works are the natural background for the small-block question. The Fabius–Rvachev construction and its smooth compactly supported density are described by Arias de Reyna [5].

The contribution here is the proved critical-complement theorem package for the moving geometric tilt: a fixed-complement phase expansion, a uniform leading crossover for every $m = o(n)$, a sharp entropy constant, and an information-loss interpretation. The Gamma rate function and Lambert W are standard tools. Nor is the general fact that a scalar constraint becomes a likelihood depending on a block sum a new principle. The value of the calculation is to identify the correct boundary profile, to control its vanishing likelihood level, and to prove the change of asymptotic regime with all normalizations specified.

There is no assertion here that a targeted literature search certifies publication priority. The mathematical claims are the displayed theorems and proofs. Their relationship to the entire conditional-limit literature is a separate priority-review question.

1.3 Conventions

All logarithms are natural. For probability measures on the same space,

$$d_{\text{TV}}(P, Q) = \sup_A |P(A) - Q(A)|, \quad D_{\text{KL}}(P \| Q) = \int \log \frac{dP}{dQ} dP.$$

The overlap is $1 - d_{\text{TV}}(P, Q) = \int \min(dP, dQ)$; there is no extra factor of two in this definition. Differential entropy is $h(g) = - \int g \log g$, with $0 \log 0 = 0$. All big- O constants may depend on fixed q and ρ . A fixed- m statement permits dependence on m as well; it is not uniform in a growing complement unless explicitly stated.

2 Model and main results

2.1 The phase convention and the observed block

Let

$$S_q = \sum_{j \geq 1} q^j U_j, \quad U_j \stackrel{\text{iid}}{\sim} \text{Unif}[0, 1], \quad 0 < q < 1. \quad (2.1)$$

Write P for the original product law. Fix $1 \leq \rho < q^{-1}$ and put

$$t_n = \rho q^{-(n+1)}, \quad T_n = t_n S_q, \quad \frac{dQ_n}{dP} = \frac{e^{-T_n}}{\mathbb{E}_P e^{-T_n}}, \quad \mu_n = \mathbb{E}_{Q_n} T_n, \quad x_n = \mu_n / t_n. \quad (2.2)$$

Our conditioned law is

$$P_n = P(\cdot \mid S_q \leq x_n). \quad (2.3)$$

The boundary starts at coordinate $n + 1$, whose scaled cap is ρ . We call the first n coordinates the bulk. This is a phase convention, not a claim that the physical transition consists of an abrupt cutoff.

Under Q_n , the variables $X_{n,j} = t_n q^j U_j$ are independent, with

$$p_a(x) = \frac{e^{-x}}{1 - e^{-a}} \mathbf{1}_{(0,a)}(x), \quad a = t_n q^j. \quad (2.4)$$

For $0 \leq m < n$, observe the prefix $B_{n,m} = \{1, \dots, n - m\}$. Let $P_{n,m}$ and $Q_{n,m}$ be the corresponding marginals of P_n and Q_n , and set

$$O_{n,m} = 1 - d_{\text{TV}}(P_{n,m}, Q_{n,m}), \quad D_{n,m} = D_{\text{KL}}(P_{n,m} \| Q_{n,m}), \quad \varepsilon_n = (2\pi n)^{-1/2}.$$

The complement contains m bulk coordinates *and the whole infinite boundary tail*. In particular, $m = 0$ does not mean observing the entire infinite configuration.

2.2 The exact complement object

On a separate product space, let the Y_ℓ be independent with density $p_{\rho q^\ell}$, and define

$$R_m = \sum_{\ell=-m}^{\infty} Y_\ell, \quad r_m = \mathbb{E} R_m, \quad Z_m = R_m + E, \quad E \sim \text{Exp}(1) \quad (2.5)$$

with E independent of the series. The complement under Q_n has exactly the law of R_m , not just an asymptotic approximation. Write g_m for the density of Z_m and $h_m = h(g_m)$. Further set

$$A_m = \frac{\rho q^{-m}}{1 - q}, \quad K_m = K_{q,\rho,m} = \log \mathbb{E} e^{R_m} = \sum_{\ell=-m}^{\infty} \log \frac{\rho q^\ell}{1 - e^{-\rho q^\ell}}. \quad (2.6)$$

Thus $0 \leq R_m \leq A_m$. All these quantities are finite for fixed m . The extra exponential variable in Z_m is essential: it arises from the inequality in (2.3).

For later refinements define

$$v(a) = 1 - \frac{a^2 e^a}{(e^a - 1)^2}, \quad (2.7)$$

$$\delta = \sum_{r=1}^{\infty} \{v(\rho q^{-r}) - 1\} + \sum_{\ell=0}^{\infty} v(\rho q^{\ell}), \quad (2.8)$$

$$\mathcal{E}_m = \frac{\delta}{2} + \frac{1}{12} - \frac{1}{2} \text{Cov}\left((Z_m - \mathbb{E}Z_m)^2, \log g_m(Z_m)\right). \quad (2.9)$$

Here $v(a)$ is the variance of p_a . The series defining δ converges; indeed $\text{Var}_{Q_n}(T_n) = n + \delta + o(1)$.

2.3 Fixed complements: phase constants and a quantitative correction

Theorem 2.1 (Fixed-complement expansion). *For fixed q, ρ, m , as $n \rightarrow \infty$,*

$$O_{n,m} = \varepsilon_n \left(\log \frac{1}{\varepsilon_n} + 1 + K_m \right) + o(\varepsilon_n), \quad (2.10)$$

$$D_{n,m} = \log \frac{1}{\varepsilon_n} - h_m + \frac{\mathcal{E}_m}{n} + O(n^{-2}). \quad (2.11)$$

More precisely, with

$$\Delta_m(a) = \int_0^{A_m} \left(1 - \frac{g_m(u)}{a} \right)_+ du, \quad (2.12)$$

one has $\Delta_m(a) \rightarrow 0$ as $a \downarrow 0$ and

$$O_{n,m} = \varepsilon_n \left(\log \frac{1}{\varepsilon_n} + 1 + K_m - \Delta_m(\varepsilon_n) \right) + O\left(\frac{\varepsilon_n (\log n)^3}{n}\right). \quad (2.13)$$

The compact integral Δ_m is an exact profile correction, not a fitted coefficient. For the dyadic example it explains why convergence to (2.10) is visibly much slower than convergence of entropy. The proof is in Section 5.

2.4 Every sublinear complement: the overlap crossover

Let $I(y) = y - 1 - \log y$ for $y > 0$. For $z > 0$, denote the two solutions of $I(y) = z$ by

$$y_-(z) = -W_0(-e^{-1-z}) < 1, \quad y_+(z) = -W_{-1}(-e^{-1-z}) > 1. \quad (2.14)$$

Theorem 2.2 (Critical-complement crossover). *Let $1 \leq m = m_n < n$ and $m = o(n)$. Put $d_{n,m} = \frac{1}{2} \log(n/m)$ and $z_{n,m} = d_{n,m}/m$. Then*

$$O_{n,m} \sim \frac{m}{\sqrt{2\pi n}} [y_+(z_{n,m}) - y_-(z_{n,m})]. \quad (2.15)$$

In particular,

$$m = o(\log n) : \quad O_{n,m} \sim \frac{\log n}{2\sqrt{2\pi n}}, \quad (2.16)$$

$$\frac{m}{\log n} \rightarrow c \in (0, \infty) : \quad O_{n,m} \sim \frac{\log n}{\sqrt{2\pi n}} C(c), \quad C(c) = c [y_+(1/(2c)) - y_-(1/(2c))], \quad (2.17)$$

$$\log n = o(m), \quad m = o(n) : \quad O_{n,m} \sim \sqrt{\frac{2}{\pi}} \sqrt{\frac{m}{n} \log \frac{n}{m}}. \quad (2.18)$$

The first assertion (2.16) also holds for $m = 0$ and for sequences that take the value zero. Moreover,

$$C(c) \longrightarrow \frac{1}{2} \quad (c \downarrow 0), \quad C(c) \sim 2\sqrt{c} \quad (c \rightarrow \infty). \quad (2.19)$$

Equation (2.15) is a leading equivalent, not a second-order expansion. In particular, expanding its Lambert roots further at fixed m does not replace the phase expansion in Theorem 2.1.

All three regimes have $d_{TV} \rightarrow 1$. What changes is the small amount of mass that the two laws still share. Section 6 proves (2.15) by treating the possible limits of $z_{n,m}$; no unproved uniform substitution into a fixed-fraction theorem is used.

2.5 Entropy and information loss

Theorem 2.3 (Entropy with a growing critical complement). *If $m \rightarrow \infty$ and $m = o(n)$, then*

$$D_{n,m} = \frac{1}{2} \log \frac{n}{m} - \frac{1}{2} + o(1). \quad (2.20)$$

For the entire infinite configuration,

$$D_{KL}(P_n \| Q_n) = \frac{1}{2} \log(2\pi n) - 1 + o(1). \quad (2.21)$$

Consequently, for each fixed m ,

$$D_{KL}(P_n \| Q_n) - D_{n,m} \longrightarrow h(R_m + E) - 1 = I(R_m; R_m + E). \quad (2.22)$$

The mutual information on the right is finite and strictly positive. Its interpretation is precise: the limiting cost of hiding the boundary complement is the information that its tilted sum transmits through independent one-sided exponential noise. The proof is in Section 7.

3 An exact likelihood and the Fabius boundary profile

3.1 Existence of the tilted product

The original series converges absolutely, since $0 \leq S_q \leq q/(1-q)$. Its Laplace transform is positive. Finite product factorization and bounded convergence therefore give the tilted product in (2.4), including independence of any finite coordinate set from its infinite tail. The mean and variance of one coordinate are

$$b(a) = 1 - \frac{a}{e^a - 1}, \quad v(a) = 1 - \frac{a^2 e^a}{(e^a - 1)^2}. \quad (3.1)$$

They follow by differentiating the integral of e^{-x} over $(0, a)$. At zero, $b(a) = a/2 + O(a^2)$ and $v(a) = a^2/12 + O(a^4)$; at infinity, $b(a) = 1 + O(ae^{-a})$ and $v(a) = 1 + O(a^2 e^{-a})$. It follows that

$$\mu_n = n + c_{q,\rho} + o(1), \quad \text{Var}_{Q_n}(T_n) = n + \delta + o(1), \quad (3.2)$$

where

$$c_{q,\rho} = r_0 - \sum_{r \geq 1} \frac{\rho q^{-r}}{e^{\rho q^{-r}} - 1}.$$

The errors in (3.2) in fact decrease faster than any inverse power of n . Also $r_m = m + O(1)$ and $\text{Var}(R_m) = m + O(1)$, uniformly in m .

3.2 The density that controls the answer

Proposition 3.1 (Exact boundary profile). *Let $F_q(x) = P(S_q \leq x)$, extended by zero and one outside its support. For $u > 0$,*

$$g_m(u) = e^{-u} \mathbb{E}[e^{R_m} \mathbf{1}_{\{R_m \leq u\}}] = e^{K_m - u} F_q\left(\frac{q^{m+1}u}{\rho}\right). \quad (3.3)$$

The density is continuous, positive on $(0, \infty)$, log-concave, bounded by one, and satisfies the exact tail identity

$$g_m(u) = e^{K_m - u} \quad \text{for } u \geq A_m. \quad (3.4)$$

Its entropy and all integrals $\int (1 + u^j) g_m(u) |\log g_m(u)| du$, $j \geq 0$, are finite.

Proof. Convolution with the independent E gives the first equality in (3.3). Before tilting, the complement sum has the law of $\rho q^{-m-1} S_q$. Its tilted law is obtained by multiplying by e^{-u} and normalizing. Since the support is bounded, reversing this tilt inside the expectation is legitimate and gives the second equality. The normalization is $\mathbb{E}e^{R_m} = e^{K_m}$, obtained by taking the product of the one-coordinate integrals $a/(1 - e^{-a})$.

Every positive interval $(0, u)$ has positive complement probability: choose a finite prefix small enough and make its deterministic remaining tail less than $u/2$. This also shows $F_q(x) > 0$ for $x > 0$. The complement has no atoms because one of its independent summands has a density. Continuity of its CDF gives continuity of g_m . The first expression in (3.3) gives $g_m \leq 1$ directly. The tail identity follows from $R_m \leq A_m$.

Each truncated exponential density, and the exponential density, is log-concave. Convolution preserves log-concavity by Prékopa's theorem [6, Theorem 7]. Apply that result to finite partial sums plus E . Their densities converge pointwise to g_m : in the first formula of (3.3), partial sums converge almost surely, the exponential factor is bounded for fixed m , and the limiting sum has no atoms. Pointwise limits preserve the log-concavity inequality. Finally, $-x \log x$ is bounded on $[0, 1]$, so the entropy integrand is integrable on $[0, A_m]$. The exact exponential tail handles every polynomially weighted integral on the remainder of the half-line. $\square$

3.3 Reduction of both distances to one dimension

Let

$$W_{n,m} = \sum_{j \in B_{n,m}} X_{n,j}, \quad V_{n,m} = W_{n,m} - \mathbb{E}_{Q_n} W_{n,m},$$

and let $f_{n,m}$ denote the density of the centered sum $V_{n,m}$. Extend this density by zero off its support. Define the common normalizer

$$a_n = \mathbb{E}_{Q_n} [e^{T_n - \mu_n} \mathbf{1}_{\{T_n \leq \mu_n\}}]. \quad (3.5)$$

It depends on n, q, ρ , but not on the chosen prefix.

Proposition 3.2 (Exact marginal likelihood). *The likelihood of the conditioned prefix relative to its tilted product law is*

$$\frac{dP_{n,m}}{dQ_{n,m}} = \frac{g_m(r_m - V_{n,m})}{a_n}. \quad (3.6)$$

Under $P_{n,m}$, the variable $u = r_m - V_{n,m}$ has density

$$p_{n,m}^*(u) = \frac{f_{n,m}(r_m - u) g_m(u)}{a_n}, \quad u > 0. \quad (3.7)$$

In particular,

$$a_n = \int_0^\infty f_{n,m}(r_m - u) g_m(u) du, \quad (3.8)$$

$$O_{n,m} = \int_0^\infty f_{n,m}(r_m - u) \min\{1, g_m(u)/a_n\} du, \quad (3.9)$$

$$D_{n,m} = -\log a_n + \int_0^\infty p_{n,m}^*(u) \log g_m(u) du. \quad (3.10)$$

Proof. Changing from P to Q_n in the conditioning event gives the full likelihood $e^{T_n - \mu_n} \mathbf{1}_{\{T_n \leq \mu_n\}} / a_n$. Given a prefix with centered sum v , the independent complement is R_m and $T_n - \mu_n = v + R_m - r_m$. Conditional expectation of the full likelihood is therefore

$$\frac{e^{v - r_m}}{a_n} \mathbb{E}[e^{R_m} \mathbf{1}_{\{R_m \leq r_m - v\}}] = \frac{g_m(r_m - v)}{a_n}.$$

This proves (3.6). A change of variable from v to $u = r_m - v$ proves (3.7) and its normalization. Integrating $\min(1, dP_{n,m}/dQ_{n,m})$ gives the overlap. Integrating the logarithm of the likelihood under $P_{n,m}$ gives the entropy formula. All formulas concern the entire prefix marginal, not merely an approximation by its sum: the likelihood is exactly a function of that sum. $\square$

Remark 3.3 (Why inequality conditioning matters). For a regular conditional law on an exact level $T_n = \mu_n$, the analogous complement factor is the density of R_m , not that of $R_m + E$. The exponential tail (3.4), the constant 1 in the overlap, and the information gap (2.22) all use the inequality in the present problem. Interchanging these two conditioning operations would change the answer.

4 Signed Gamma convolutions and local estimates

4.1 An exact algebraic representation

Write

$$\gamma_k(x) = \frac{e^{-x} x^{k-1}}{\Gamma(k)} \mathbf{1}_{(0, \infty)}(x)$$

for the unit-rate Gamma density. Put $a_r = \rho q^{-r}$ for $r \geq 1$ and introduce the finite signed measure

$$\nu_a = \frac{\delta_0 - e^{-a} \delta_a}{1 - e^{-a}}. \quad (4.1)$$

The elementary identity

$$p_a = \gamma_1 * \nu_a \quad (4.2)$$

is valid pointwise away from endpoints. Although ν_a is signed, the convolution on the right is the nonnegative truncated exponential density. For $k = n - m$,

$$\mathcal{L}(W_{n,m}) \text{ has density } \gamma_k * \nu_{m,n}, \quad \nu_{m,n} = \nu_{a_{m+1}} * \cdots * \nu_{a_n}. \quad (4.3)$$

This is an exact identity for the actual geometric model.

Lemma 4.1 (Uniform control of the signed correction). *For every $0 \leq \theta < 1$ there is a finite C_θ , depending only on q, ρ, θ , such that*

$$\int e^{\theta u} |\nu_{m,n}|(du) \leq C_\theta \quad (0 \leq m < n). \quad (4.4)$$

Every $\nu_{m,n}$ has total mass one and support in $[0, \infty)$. For fixed m it converges in these weighted total-variation norms to $\nu_{m,\infty}$. The error decreases faster than any inverse power of n . Also $\nu_{m,\infty} \rightarrow \delta_0$ in the same norms as $m \rightarrow \infty$.

Proof. The weighted variation of a convolution is at most the product of the weighted variations. Hence the left side of (4.4) is at most

$$\prod_{r=m+1}^n \frac{1 + e^{-(1-\theta)a_r}}{1 - e^{-a_r}} \leq \prod_{r=1}^{\infty} \frac{1 + e^{-(1-\theta)a_r}}{1 - e^{-a_r}} < \infty.$$

Convergence follows because a_r grows geometrically and $\sum_r e^{-ca_r} < \infty$ for every $c > 0$. More explicitly, $v_a - \delta_0 = e^{-a}(\delta_0 - \delta_a)/(1 - e^{-a})$. Telescoping the convolution product bounds the tail norm by a constant multiple of $\sum_{r>N} e^{-(1-\theta)a_r}$. This proves all convergence claims, including the stated speed. Mass and support follow directly from (4.1). $\square$

The signed representation is only an analytic device. No probabilistic argument treats a signed measure as the law of a random variable.

4.2 Central density estimates and their first correction

Define

$$d_m = \sum_{r=m+1}^{\infty} \{v(a_r) - 1\}, \quad B_m(v) = \frac{m}{2} - \frac{v^2}{2} - v - \frac{d_m}{2} - \frac{1}{12}. \quad (4.5)$$

Lemma 4.2 (Prefix density estimates). *Suppose $m = o(n)$. Then*

$$\sup_v f_{n,m}(v) \leq C\varepsilon_n, \quad (4.6)$$

and for every sequence $b_n = o(\sqrt{n})$,

$$\sup_{|v| \leq b_n} \left| \frac{f_{n,m}(v)}{\varepsilon_n} - 1 \right| \rightarrow 0. \quad (4.7)$$

If m is fixed, then for every fixed C_0 , uniformly for $|v| \leq C_0 \log n$,

$$f_{n,m}(v) = \varepsilon_n \left[1 + \frac{B_m(v)}{n} + O\left(\frac{1 + |v|^6}{n^2}\right) \right]. \quad (4.8)$$

Proof. Stirling's formula gives $\sup_x \gamma_k(x) \leq C/\sqrt{k}$. Combine it with (4.3) and (4.4) to obtain (4.6). The mean shift of the signed correction is

$$\beta_{m,n} = \int u v_{m,n}(du) = - \sum_{r=m+1}^n \frac{a_r}{e^{a_r} - 1},$$

which is bounded uniformly. Thus

$$f_{n,m}(v) = \int \gamma_k(k + \beta_{m,n} + v - u) v_{m,n}(du).$$

For $|w| = o(\sqrt{k})$, uniformly, $\sqrt{2\pi k} \gamma_k(k + w) \rightarrow 1$. Restrict the signed integral to $u \leq h_n$, where $h_n \rightarrow \infty$ and $h_n = o(\sqrt{n})$, and use the exponential variation bound on the rest. Its signed mass is one. Since $k/n \rightarrow 1$, this proves (4.7).

For the refinement, Stirling's formula and the Taylor expansion of $\log(1 + w/k)$ give, uniformly on logarithmic windows,

$$\gamma_k(k + w) = \frac{1}{\sqrt{2\pi k}} \left[1 - \frac{w^2/2 + w + 1/12}{k} + O\left(\frac{1 + |w|^6}{k^2}\right) \right]. \quad (4.9)$$

Integrate this expansion against the signed correction. To justify the integration of the remainder, first restrict to $u \leq C_1 \log n$ with C_1 sufficiently large. The excluded weighted variation is smaller than any prescribed inverse power of n , by (4.4); all remaining polynomial moments are uniformly bounded. The first two signed centered moments are zero and $\text{Var}(W_{n,m}) - k = d_m + o(n^{-N})$ for every fixed N . Consequently, after centering,

$$\int w \, d\nu_{m,n} = v, \quad \int w^2 \, d\nu_{m,n} = v^2 + d_m + o(n^{-N}).$$

Finally $(2\pi k)^{-1/2} = \varepsilon_n [1 + m/(2n) + O(n^{-2})]$. Substitution in (4.9) proves (4.8). $\square$

Corollary 4.3 (Normalization to two orders). *With δ from (2.8),*

$$a_n = \varepsilon_n \left[1 - \frac{\delta/2 + 1/12}{n} + O(n^{-2}) \right]. \quad (4.10)$$

In particular $a_n/\varepsilon_n \rightarrow 1$.

Proof. Use (3.8) with $m = 0$ and apply (4.8). The exponential tail of g_0 permits restriction to $u \leq C \log n$ with an error of arbitrarily high inverse-polynomial order, using the global bound on f . All required polynomial moments are finite. More generally, for any fixed m , if Z_m has density g_m , then

$$\mathbb{E}(r_m - Z_m) = -1, \quad \mathbb{E}(r_m - Z_m)^2 = \text{Var}(R_m) + 2,$$

so

$$\mathbb{E}B_m(r_m - Z_m) = -\frac{\delta}{2} - \frac{1}{12}, \quad (4.11)$$

because $d_m + \text{Var}(R_m) - m = \delta$. Integrating the polynomial remainder in (4.8) gives $O(n^{-2})$. $\square$

4.3 A uniform saddle estimate for the complement

Put

$$\mathcal{H}_m(s) = \mathbb{E}e^{sR_0} \prod_{r=1}^m \frac{1 - e^{-(1-s)a_r}}{1 - e^{-a_r}}, \quad \Re s < 1. \quad (4.12)$$

The product converges locally uniformly, together with all derivatives, to an analytic function $\mathcal{H}(s)$. It is strictly positive for real $s < 1$ and $\mathcal{H}(0) = 1$. Directly from the truncated exponential integrals,

$$\mathbb{E}e^{sZ_m} = (1-s)^{-(m+1)} \mathcal{H}_m(s), \quad \Re s < 1. \quad (4.13)$$

Lemma 4.4 (Complement saddle estimate). *For any compact interval $J \subset (0, \infty)$, uniformly for $y \in J$,*

$$g_m(my) = \frac{\mathcal{H}(1-1/y)}{\sqrt{2\pi m}} e^{-mI(y)} [1 + O_J(m^{-1})], \quad m \rightarrow \infty. \quad (4.14)$$

Also, uniformly for y in a compact interval on either side of one,

$$\mathbb{P}(Z_m \geq my) \leq C_J e^{-mI(y)} \quad (y > 1), \quad (4.15)$$

$$\mathbb{P}(Z_m \leq my) \leq C_J e^{-mI(y)} \quad (y < 1). \quad (4.16)$$

The constants in these tail bounds remain bounded as $y \rightarrow 1$.

Proof. Let $\eta_m = \mathcal{L}(R_0) * \nu_{0,m}$. The signed convolution identity gives

$$g_m = \gamma_{m+1} * \eta_m. \quad (4.17)$$

The measures η_m have support in $[0, \infty)$, mass one, and uniformly bounded exponentially weighted variations for every exponent less than one. Their moment generating functions are $\mathcal{H}_m$. For $u \geq 0$,

$$\frac{\gamma_{m+1}(my - u)}{\gamma_{m+1}(my)} = e^u \left(1 - \frac{u}{my}\right)^m \mathbf{1}_{\{u < my\}}. \quad (4.18)$$

If $s = 1 - 1/y$, this converges to e^{su} . Moreover, $\log(1 - z) \leq -z$ bounds the ratio by e^{su} . Choose $\theta < 1$ strictly larger than every s arising from J . The exponential variation bound makes the integration uniform in y . On $u \leq my/2$, Taylor's theorem bounds the difference from e^{su} by $C_J m^{-1} u^2 e^{su}$, since $1 - e^{-v} \leq v$ for $v \geq 0$. On $u > my/2$ the stronger exponential moment makes the integral exponentially small in m . Therefore

$$\int \frac{\gamma_{m+1}(my - u)}{\gamma_{m+1}(my)} \eta_m(du) = \mathcal{H}_m(1 - 1/y) + O_J(m^{-1}) = \mathcal{H}(1 - 1/y) + O_J(m^{-1}).$$

Stirling's formula yields

$$\gamma_{m+1}(my) = \frac{e^{-mI(y)}}{\sqrt{2\pi m}} [1 + O(m^{-1})].$$

Positivity and continuity of $\mathcal{H}$ on the relevant compact interval allow the error to be written relatively, proving (4.14).

For the tails, apply the exponential Markov inequality to (4.13) with $s = 1 - 1/y$, positive or negative as appropriate. Its exponential part is $e^{-mI(y)}$; the remaining factor is $y\mathcal{H}_m(1 - 1/y)$, uniformly bounded on J , also near one. $\square$

Remark 4.5 (Why a central limit theorem alone is insufficient). The overlap is determined by the region $g_m(u) \gtrsim n^{-1/2}$. When $m \asymp \log n$, this is an exponentially small density level relative to the complement's central scale $m^{-1/2}$. The relevant points are a distance of order m , not $\sqrt{m}$, from the center. The rate function in (4.14), rather than its quadratic Taylor polynomial alone, determines the answer.

5 Fixed complements: proof of the phase expansions

5.1 An exact clipped-density integral

For a fixed complement, put

$$J_m(a) = \int_0^\infty \min\{1, g_m(u)/a\} du.$$

Lemma 5.1 (Exact boundary-deficit formula). *For $0 < a < e^{K_m - A_m}$,*

$$J_m(a) = \log(1/a) + K_m + 1 - \Delta_m(a). \quad (5.1)$$

Moreover $\Delta_m(a) \rightarrow 0$ as $a \downarrow 0$. If a, b both satisfy the same upper bound, then

$$|J_m(a) - J_m(b)| \leq (A_m + 1) |\log(a/b)|. \quad (5.2)$$

Proof. The tail crossing occurs at $u = \log(e^{K_m}/a) > A_m$. By (3.4),

$$\int_{A_m}^{\infty} \min\{1, g_m(u)/a\} du = \log(e^{K_m}/a) - A_m + 1.$$

On $[0, A_m]$, the integral is $A_m - \Delta_m(a)$. This proves (5.1). Since $g_m(u) > 0$ for $u > 0$, bounded convergence on this compact interval proves $\Delta_m(a) \rightarrow 0$. For fixed $g \geq 0$, the function $v \mapsto \min(1, ge^{-v})$ is 1-Lipschitz. Its contribution on $[0, A_m]$ is thus A_m -Lipschitz in $v = \log a$. The explicit tail integral has slope -1 in v . Adding the two estimates gives (5.2). $\square$

5.2 Overlap, including the controlled correction

Proof of (2.13) and (2.10). Fix m . On $0 \leq u \leq C \log n$, Lemma 4.2 gives

$$f_{n,m}(r_m - u) = \varepsilon_n \left[1 + O\left(\frac{1+u^2}{n}\right) \right].$$

Since $a_n \asymp \varepsilon_n$ and $g_m(u) = e^{K_m - u}$ beyond A_m ,

$$\int_0^{C \log n} (1+u^2) \min\{1, g_m(u)/a_n\} du = O((\log n)^3).$$

The global bound $f \leq C\varepsilon_n$ and the exact exponential tail show that the integral beyond $C \log n$ is smaller than ε_n/n^2 when C is sufficiently large. Apply these facts to (3.9) to obtain

$$O_{n,m} = \varepsilon_n J_m(a_n) + O\left(\frac{\varepsilon_n (\log n)^3}{n}\right).$$

Corollary 4.3 gives $\log(a_n/\varepsilon_n) = O(n^{-1})$, so (5.2) permits replacing a_n by ε_n in J_m . The exact formula (5.1) now proves (2.13). Taking $\Delta_m(\varepsilon_n) \rightarrow 0$ proves (2.10). $\square$

The $+1$ in the constant comes from integrating the exponential tail past the likelihood crossing. The K_m term is its amplitude. The small negative correction $-\Delta_m$ comes from the other end, where the compactly supported uniform-series distribution is exceptionally flat near zero. These are distinct mechanisms.

5.3 Relative entropy through order $1/n$

Proof of (2.11). Let $Z = Z_m$, $g = g_m$, and put $b_* = -\delta/2 - 1/12$. Combining (4.8), (4.10), and (3.7) gives, in integrals against $g(u)$,

$$p_{n,m}^*(u) = g(u) \left[1 + \frac{B_m(r_m - u) - b_*}{n} + O\left(\frac{1+u^6}{n^2}\right) \right]. \quad (5.3)$$

Here and below an integral expansion is justified first on a logarithmic window. Outside that window, the global bound on f/a_n and the exact exponential tail of g make the error smaller than any required inverse power of n . Proposition 3.1 also makes $(1+u^6)g(u)|\log g(u)|$ integrable. Thus (5.3) may be integrated against $\log g$ with error $O(n^{-2})$.

The centered polynomial in (5.3) is

$$B_m(r_m - u) - b_* = -\frac{1}{2} [(u - r_m - 1)^2 - \text{Var}(Z_m)].$$

Consequently,

$$\int p_{n,m}^* \log g = -h_m - \frac{1}{2n} \text{Cov}\left((Z_m - \mathbb{E}Z_m)^2, \log g_m(Z_m)\right) + O(n^{-2}).$$

Finally,

$$-\log a_n = \log(1/\varepsilon_n) + \frac{\delta/2 + 1/12}{n} + O(n^{-2}).$$

Substitute these two expansions into (3.10). This is (2.11) with the coefficient in (2.9). $\square$

Remark 5.2 (The two fixed-complement remainders are different). The entropy expansion has an ordinary inverse-power remainder because its integral weights the small-density region by g_m . Overlap clips the likelihood at one. This retains the geometric length of a small boundary interval and produces $\Delta_m(\varepsilon_n)$, which need not be an inverse power of n . Omitting this distinction leads to misleading finite-size extrapolations.

6 The logarithmic overlap transition

We now prove Theorem 2.2. Write $m = m_n$, $d = \frac{1}{2} \log(n/m)$, and $z = d/m$ when $m \geq 1$. Throughout this section $m = o(n)$, so $d \rightarrow \infty$, and $a_n \sim \varepsilon_n$.

6.1 The general window comparison

Suppose an interval $L_n \subset (0, \infty)$ is such that $|r_m - u| = o(\sqrt{n})$ uniformly on L_n . By (4.7),

$$\int_{L_n} f_{n,m}(r_m - u) \min\{1, g_m(u)/a_n\} du = (1 + o(1))\varepsilon_n \int_{L_n} \min\{1, g_m(u)/a_n\} du. \quad (6.1)$$

The contribution of its complement can be bounded without any relative local approximation:

$$\int_{L_n^c} f_{n,m}(r_m - u) \min\{1, g_m(u)/a_n\} du \leq \mathbb{C}\mathbb{P}(Z_m \notin L_n). \quad (6.2)$$

This follows from $f \leq C\varepsilon_n$ and $a_n \asymp \varepsilon_n$. The usefulness of (6.2) is that an ordinary Chernoff bound suffices beyond a slightly enlarged level-set interval.

6.2 Thin complements: $m = o(\log n)$

Each truncated exponential is stochastically bounded above by a full unit exponential. Moreover $R_0 \leq A_0$. Hence, using independent couplings,

$$R_m \leq_{\text{st}} \Gamma(m, 1) + A_0, \quad Z_m \leq_{\text{st}} \Gamma(m + 1, 1) + A_0. \quad (6.3)$$

Here the Gamma variable of shape zero is interpreted as zero. Since $\mathbb{E}R_m \leq m + A_0$, Markov's inequality and (3.3) give

$$g_m(u) \geq \frac{1}{2}e^{-u} \quad \text{for } u \geq 2(m + A_0). \quad (6.4)$$

Fix $0 < \eta < 1/2$. On the interval

$$2(m + A_0) \leq u \leq (1/2 - \eta) \log n,$$

which has length $(1/2 - \eta + o(1)) \log n$, one has $g_m(u)/a_n \rightarrow \infty$ uniformly. The prefix density is flat there because $r_m = O(m + 1)$ and $\log n = o(\sqrt{n})$. Thus

$$\liminf \frac{O_{n,m}}{\varepsilon_n \log n} \geq \frac{1}{2} - \eta.$$

For the upper bound, the part $u \leq (1/2 + \eta) \log n$ contributes at most $(1/2 + \eta + o(1))\varepsilon_n \log n$. For the rest, choose a fixed $\theta \in (0, 1)$ with $\theta(1/2 + \eta) > 1/2$. By (6.3),

$$\mathbb{P}(Z_m > (1/2 + \eta) \log n) \leq e^{\theta A_0} (1 - \theta)^{-(m+1)} n^{-\theta(1/2 + \eta)} = o(\varepsilon_n \log n).$$

Equation (6.2) controls the tail. Letting $\eta \downarrow 0$ proves (2.16), including $m = 0$.

6.3 The critical window: $z \rightarrow z_0 \in (0, \infty)$

Now $m \rightarrow \infty$ and $m = O(\log n)$. The saddle estimate gives, uniformly on fixed compact intervals of positive y ,

$$\log \frac{g_m(my)}{a_n} = d - mI(y) + O(1). \quad (6.5)$$

Fix $0 < \eta < 1$. The interval between the two roots of $I(y) = (1 - \eta)z$ lies inside the set $g_m(my) \geq a_n$ for all large n . Its endpoints remain in a compact subset of $(0, \infty)$. Similarly, let L_n be the interval between the roots of $I(u/m) = (1 + \eta)z$. On this interval the prefix density is flat, since $u = O(m) = O(\log n)$. Outside it, Lemma 4.4 gives

$$\mathbb{P}(Z_m \notin L_n) \leq C e^{-(1+\eta)d}.$$

Relative to the anticipated scale $\varepsilon_n m$, this bound is negligible:

$$\frac{e^{-(1+\eta)d}}{\varepsilon_n m} = O\left(m^{-1/2} e^{-\eta d}\right) \rightarrow 0.$$

Equations (6.1)–(6.2) therefore bound $O_{n,m}/(\varepsilon_n m)$ between the widths of the two root intervals, up to $o(1)$. Let $\eta \downarrow 0$. Continuity of the roots proves

$$O_{n,m} \sim \varepsilon_n m [y_+(z) - y_-(z)] \quad \text{when } z \rightarrow z_0 \in (0, \infty). \quad (6.6)$$

If $m/\log n \rightarrow c \in (0, \infty)$, then $z \rightarrow 1/(2c)$, giving (2.17).

6.4 The Gaussian side: $z \rightarrow 0$

In this regime $d = o(m)$, though still $d \rightarrow \infty$. Set

$$L = \sqrt{2md} = \sqrt{m \log(n/m)}.$$

For fixed $\eta \in (0, 1)$, Taylor's formula for I yields

$$mI(1 \pm (1 \pm \eta)L/m) = (1 \pm \eta)^2 d + o(d).$$

The two signs in this expression indicate the left/right endpoints and the inner/outer enlargement separately. By (6.5), the inner interval $[m - (1 - \eta)L, m + (1 - \eta)L]$ lies in $g_m \geq a_n$ for large n . For the outer interval with $1 + \eta$, Chernoff's inequality gives

$$\mathbb{P}(|Z_m - m| > (1 + \eta)L) \leq C \exp\{-(1 + \eta)^2 d + o(d)\}.$$

Compared with $\varepsilon_n L$, this is bounded by a constant times

$$\frac{\exp\{-(1 + \eta)^2 d + o(d)\}}{\sqrt{d}} \rightarrow 0.$$

The local flatness needed on the outer interval holds, because

$$\frac{L^2}{n} = \frac{m}{n} \log \frac{n}{m} \rightarrow 0.$$

The interval comparison thus gives $O_{n,m} \sim 2\varepsilon_n L$. Since $y_+(z) - y_-(z) \sim 2\sqrt{2z}$ as $z \downarrow 0$, this is (2.15) in the regime $z \rightarrow 0$, as well as (2.18) under its stated hypotheses.

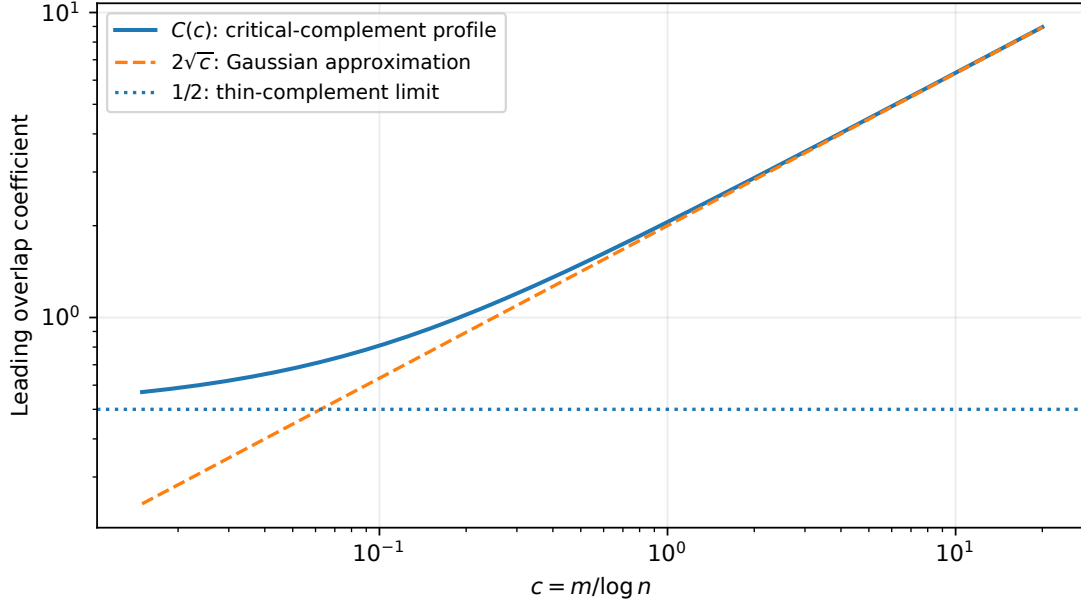


Figure 1: The critical coefficient $C(c)$ in (2.17). The Gaussian expression $2\sqrt{c}$ is its large- c limit, not its value throughout the critical window. The constant $1/2$ is the thin-complement limit. The curves are deterministic evaluations of the theorem's Lambert- W formula, not Monte Carlo estimates.

6.5 Completion of the uniform leading formula

If $z \rightarrow \infty$, then $m = o(\log n)$, since $mz = d \leq \frac{1}{2} \log n$. The elementary root asymptotics are

$$y_-(z) \sim e^{-1-z}, \quad y_+(z) \sim z, \quad y_+(z) - y_-(z) \sim z.$$

Hence $m[y_+(z) - y_-(z)] \sim d \sim \frac{1}{2} \log n$. The thin-complement result proves (2.15) in this regime.

For an arbitrary sequence $m = o(n)$ with $m \geq 1$, any subsequence has a further subsequence along which z converges in the compactified interval $[0, \infty]$. The three arguments above show that the ratio in (2.15) tends to one on every such further subsequence. Therefore the ratio tends to one on the original sequence. This proves the full assertion without assuming a limit for $m/\log n$. The same root asymptotics, and the quadratic expansion at one, give (2.19). Theorem 2.2 is proved.

7 Entropy for growing complements and the information gap

7.1 A density and entropy limit for the complement

Define the standardized density

$$b_m(z) = \sqrt{m} g_m(m + \sqrt{m} z), \tag{7.1}$$

extended by zero when its argument is nonpositive. From (4.14), $\mathcal{H}(0) = 1$, and Taylor's formula for I ,

$$b_m(z) \longrightarrow \phi(z) = (2\pi)^{-1/2} e^{-z^2/2} \tag{7.2}$$

locally uniformly on $\mathbb{R}$.

Lemma 7.1 (Uniform entropy integrability). *There are $C, c > 0$ such that, for all sufficiently large m and all z ,*

$$b_m(z) \leq C e^{-c|z|}, \quad |b_m(z) \log b_m(z)| \leq C e^{-c|z|/2}. \tag{7.3}$$

Consequently,

$$h_m = \frac{1}{2} \log(2\pi e m) + o(1). \quad (7.4)$$

Proof. The signed Gamma representation (4.17) and its bounded total variation give $\sup_z b_m(z) \leq C$. Log-concavity of g_m gives log-concavity of b_m . By (7.2), the ratios $b_m(2)/b_m(0)$ and $b_m(-2)/b_m(0)$ are bounded strictly below one for large m . Concavity of $\log b_m$ then bounds its secant slopes outside $[-2, 2]$ away from zero in the inward direction. Extrapolating these slopes proves the first inequality in (7.3); the global bound handles the central interval. Zeros of the density cause no problem, since the asserted upper bound is then automatic.

For $0 < x \leq 1$, $x|\log x| \leq C\sqrt{x}$. For $1 \leq x \leq C$, the function $x \log x$ is bounded; the exponential envelope restricts this case to a bounded interval. This proves the second inequality. Dominated convergence in (7.2) gives

$$-\int b_m \log b_m \longrightarrow -\int \phi \log \phi = \frac{1}{2} \log(2\pi e).$$

The entropy change under scaling by $\sqrt{m}$ is $\frac{1}{2} \log m$, proving (7.4). $\square$

7.2 Entropy after conditioning

Proof of (2.20). Under the density $p_{n,m}^*$ in (3.7), change variables $u = m + \sqrt{m} z$. The resulting density is

$$b_m(z) \frac{f_{n,m}(r_m - m - \sqrt{m} z)}{a_n}. \quad (7.5)$$

The ratio in (7.5) is uniformly bounded by (4.6) and $a_n \asymp \varepsilon_n$. On any fixed compact interval of z , it tends uniformly to one, because $r_m - m = O(1)$ and $\sqrt{m} = o(\sqrt{n})$. Split

$$\log g_m(m + \sqrt{m} z) = -\frac{1}{2} \log m + \log b_m(z).$$

The first term integrates exactly to $-\frac{1}{2} \log m$ since (7.5) is a probability density. This exact normalization avoids multiplying an uncontrolled convergence error by $\log m$. For the second term, Lemma 7.1 and the bounded ratio permit dominated convergence. Therefore

$$\int p_{n,m}^* \log g_m = -\frac{1}{2} \log m + \int \phi \log \phi + o(1) = -\frac{1}{2} \log(2\pi e m) + o(1).$$

Finally $-\log a_n = \frac{1}{2} \log(2\pi n) + o(1)$. Substitution into (3.10) proves $D_{n,m} = \frac{1}{2} \log(n/m) - \frac{1}{2} + o(1)$. $\square$

7.3 Full-configuration entropy and a channel identity

Proof of (2.21) and (2.22). Let $G_n = \mu_n - T_n \geq 0$ under P_n , and write F_n for the centered density of $T_n - \mu_n$ under Q_n . Convolving the prefix density for $m = 0$ with the bounded complement shows

$$\sup_v F_n(v) \leq C\varepsilon_n, \quad F_n(v)/\varepsilon_n \longrightarrow 1$$

uniformly on fixed compact intervals. The slack density under P_n is

$$\frac{e^{-g} F_n(-g)}{a_n} \mathbf{1}_{\{g \geq 0\}}.$$

It converges pointwise to e^{-g} and is bounded by Ce^{-g} . Dominated convergence with weight $1 + g$ gives $\mathbb{E}_{P_n} G_n \rightarrow 1$. The full likelihood yields the exact entropy identity

$$D_{\text{KL}}(P_n \| Q_n) = -\log a_n - \mathbb{E}_{P_n} G_n,$$

proving (2.21).

Subtract the fixed-complement formula (2.11); the limit is $h_m - 1$. Since $Z_m = R_m + E$ with E independent, the standard entropy identity for this additive channel is

$$I(R_m; Z_m) = h(Z_m) - h(Z_m | R_m) = h_m - h(E) = h_m - 1.$$

All quantities are finite by Proposition 3.1; equivalently, the identity follows directly by integrating the log ratio of the conditional density $e^{-(z-r)} \mathbf{1}_{z \geq r}$ and $g_m(z)$. It is nonnegative by relative entropy. It is strictly positive here: zero mutual information would make R_m and $R_m + E$ independent, whereas $\text{Cov}(R_m, R_m + E) = \text{Var}(R_m) > 0$. This completes Theorem 2.3. $\square$

8 Phase structure, endpoint scales, and examples

8.1 Which constants remember the phase?

Proposition 8.1 (Phase identities). *For fixed m , $K_{q,\rho,m}$ and h_m are continuous in $\rho > 0$. The overlap constant satisfies*

$$\rho \frac{\partial K_{q,\rho,m}}{\partial \rho} = r_m > 0. \quad (8.1)$$

At the phase boundary, with parameters extended continuously,

$$R_m|_{\rho=q^{-1}} \stackrel{d}{=} R_{m+1}|_{\rho=1}, \quad (8.2)$$

and the corresponding K and h constants agree. At fixed q, ρ , $h_{m+1} > h_m$ and

$$K_{m+1} - K_m = \log \frac{\rho q^{-(m+1)}}{1 - e^{-\rho q^{-(m+1)}}}. \quad (8.3)$$

Proof. The series for K_m and its derivative converge locally uniformly in ρ , using the small-cap expansion in the tail. Differentiating one term with cap a gives

$$a \frac{d}{da} \log \frac{a}{1 - e^{-a}} = 1 - \frac{a}{e^a - 1} = b(a).$$

Summing yields (8.1). Continuity of h_m follows from (3.3) and dominated convergence: on compact ρ -ranges the supports A_m and amplitudes e^{K_m} are bounded, the entropy integrand on a fixed compact interval is bounded, and the remaining tail has a common exponential envelope. The CDF F_q is continuous.

Reindexing the caps gives (8.2) and (8.3). For entropy monotonicity, Z_{m+1} is the sum of Z_m and an independent nondegenerate truncated exponential $Y_{-(m+1)}$. Therefore

$$h_{m+1} - h_m = I(Y_{-(m+1)}; Z_m + Y_{-(m+1)}) > 0.$$

Strict positivity follows as before from a nonzero covariance and finite second moments. $\square$

The phase identity also checks the indexing convention. The same tilt written with $\rho = q^{-1}$ and bulk index n is written with $\rho = 1$ and index $n + 1$. To observe the same prefix one must replace m by $m + 1$. The constants then match, while replacing n by $n + 1$ changes the fixed-complement expansions only at smaller orders.

The leading crossover (2.15) forgets q and ρ . The fixed-complement constant does not: (8.1) proves strict phase dependence of the overlap correction, not merely a numerical suggestion. We have not proved monotonicity of h_m in ρ ; its continuity and exact formula suffice for the results here.

8.2 Returning to the original small threshold

Let $L_n = \log(1/x_n)$ and $a = \log(1/q)$. Equation (3.2) gives

$$x_n \sim \frac{nq^{n+1}}{\rho}, \quad L_n = an - \log n + O(1), \quad n = \frac{L_n + \log L_n}{a} + O(1). \quad (8.4)$$

Consequently the crossover $m \asymp \log n$ occurs at

$$m \asymp \log \log(1/x_n) \quad (8.5)$$

in the original endpoint parameter. This scale is much smaller than the effective dimension, which is of order $\log(1/x_n)$. For fixed m , the two leading endpoint formulas become

$$O_{n,m} = \sqrt{\frac{a}{2\pi L_n}} \left[\frac{1}{2} \log L_n + \frac{1}{2} \log \frac{2\pi}{a} + 1 + K_m \right] + o(L_n^{-1/2}), \quad (8.6)$$

$$D_{n,m} = \frac{1}{2} \log L_n + \frac{1}{2} \log \frac{2\pi}{a} - h_m + o(1). \quad (8.7)$$

These statements are along the fixed-phase sequence in (2.2). They do not suppress the phase by replacing it with an averaged constant.

8.3 Four illustrative complement scales

For $m = 0$, the observer sees every bulk coordinate but not the infinite boundary tail. Equations (2.10) and (2.11) give $O \asymp \log n / \sqrt{n}$ and $D \sim \frac{1}{2} \log n$. The finite entropy loss relative to seeing everything is $h_0 - 1$.

For $m \sim \log n$, the overlap is $C(1) \log n / \sqrt{2\pi n}$, where numerical evaluation gives $C(1) \approx 2.055967111$. Using the Gaussian value 2 would miss a nonvanishing proportion of the leading coefficient. Entropy is already $\frac{1}{2} \log(n/\log n) - \frac{1}{2} + o(1)$.

For $m \sim n^\alpha$, $0 < \alpha < 1$,

$$O_{n,m} \sim \sqrt{\frac{2(1-\alpha)}{\pi}} n^{(\alpha-1)/2} \sqrt{\log n},$$

$$D_{n,m} = \frac{1-\alpha}{2} \log n - \frac{1}{2} + o(1).$$

For $m \sim n/(\log n)^\beta$, $\beta > 0$,

$$O_{n,m} \sim \sqrt{\frac{2\beta}{\pi}} (\log n)^{-\beta/2} \sqrt{\log \log n},$$

$$D_{n,m} = \frac{\beta}{2} \log \log n - \frac{1}{2} + o(1).$$

The final example makes the distinction between the two divergence parameters especially clear: the total-variation overlap is small but entropy may diverge only at a doubly logarithmic rate.

8.4 A warning about nonuniform fixed-complement formulas

For growing m , (8.3) gives

$$K_m = \frac{\log(1/q)}{2} m(m+1) + m \log \rho + O(1). \quad (8.8)$$

Meanwhile A_m grows geometrically in m . Thus the fixed- m tail crossing condition $\log(e^{K_m}/\varepsilon_n) > A_m$ can fail badly for a moving m . Substituting (8.8) into (2.10) is not legitimate. For example, if m is of order $\sqrt{\log n}$, the putative K_m correction is of order $\log n$, yet Theorem 2.2 still gives the thin-complement coefficient $1/2$. The profile has not reached its eventual pure exponential tail at the likelihood level being tested. This is exactly the sort of nonuniformity that the crossover theorem resolves.

9 Reproducible numerical diagnostics

9.1 What is computed

The accompanying Python code computes the boundary density through (3.3), using Fourier inversion of the *unconditioned* geometric uniform CDF. It then evaluates the exact one-dimensional likelihood formulas of Proposition 3.2. The prefix density is obtained by the signed Gamma formula (4.3). This is a direct calculation for the geometric model, not a finite-simplex proxy and not a Monte Carlo experiment.

There are still numerical approximations. The CDF is evaluated on a finite Fourier grid with a geometrically truncated product, followed by cubic interpolation. Prefix caps at least 64 are replaced by unbounded exponentials. For the dyadic $\rho = 1$ case the sum of the omitted truncation probabilities is about 1.604×10^{-28} . This is an unconditional coupling bound; it is not, by itself, an interval-certified bound for every conditional entropy integral. Adaptive quadrature and floating-point evaluation introduce further errors. None of these computations is used in the proofs.

9.2 Boundary constants

Table 1 gives representative dyadic constants. The entropy gap is $h_m - 1$, in natural-logarithm units. The $m = 0$ row at $\rho = 1$ has

$$K_0 \approx 0.944809318, \quad h_0 \approx 1.265596223, \quad \mathcal{E}_0 \approx 0.632308464.$$

For the same phase, $\delta \approx -0.9428328015$. These values are computed independently from the formulas for the variance constant, boundary profile, and covariance coefficient; the coefficient is not fitted from the finite- n table.

ρ	m	K_m	h_m	$I(R_m; R_m + E)$
1.00	0	0.944809318	1.265596223	0.265596223
1.00	1	1.783369956	1.466444743	0.466444743
1.00	2	3.188149764	1.710941221	0.710941221
1.25	0	1.164077910	1.321790468	0.321790468
1.50	0	1.376813768	1.373916185	0.373916185
1.75	0	1.583182559	1.422082460	0.422082460

Table 1: Floating-point boundary constants for $q = 1/2$. The infinite boundary tail is included in every row.

9.3 Finite-size tests of the constants

In Table 2, define

$$R_n^{\text{TV}} = \frac{O_{n,0}}{\varepsilon_n} - \log(1/\varepsilon_n) - 1 - K_0, \quad R_n^{\text{KL}} = D_{n,0} - \log(1/\varepsilon_n) + h_0.$$

Theorems predict $R_n^{\text{TV}} \rightarrow 0$ and $nR_n^{\text{KL}} \rightarrow \mathcal{E}_0 \approx 0.632308464$. The negative overlap residual is mostly $-\Delta_0(\varepsilon_n)$, as shown in Figure 2.

n	$O_{n,0}$	R_n^{TV}	$D_{n,0}$	nR_n^{KL}
32	0.294593692	-0.419386	1.406715454	0.656166
128	0.174727607	-0.334617	2.084345472	0.638468
512	0.100696423	-0.271558	2.773742630	0.633859
2048	0.056852741	-0.226850	3.465960737	0.632697
8192	0.031624816	-0.194353	4.158876182	0.632406
32768	0.017394996	-0.169404	4.851965462	0.632333

Table 2: Direct geometric-model diagnostics for $q = 1/2$, $\rho = 1$, $m = 0$. Theorem 2.1 concerns the limits, not an assertion that the two-term overlap expansion is already highly accurate at these sizes.

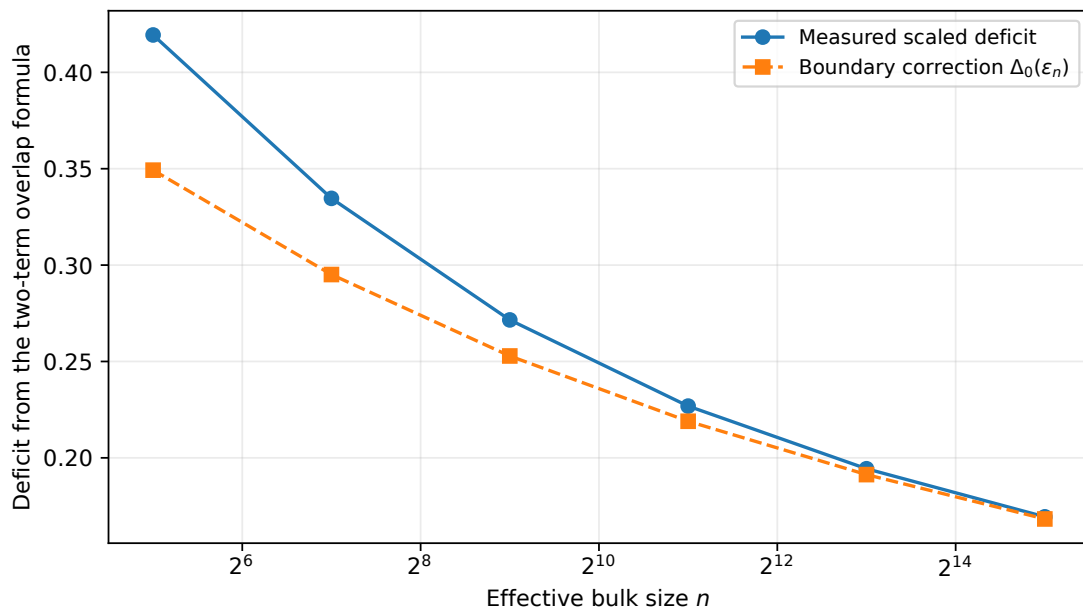


Figure 2: The measured scaled overlap deficit and the independently computed compact correction $\Delta_0(\varepsilon_n)$. Their difference is consistent with the $O((\log n)^3/n)$ error in (2.13). This plot diagnoses a slow boundary effect; it is not a proof of the asymptotic bound.

9.4 Recorded checks and reproduction

The delivered run used Python 3.13.5, NumPy 2.3.5, and SciPy 1.17.0. Across the twelve boundary computations, increasing the Fourier grid from 2^{15} to 2^{16} points changed the computed entropy by at most 1.27×10^{-14} , and the largest mass discrepancy was 6.44×10^{-15} . The residuals in the Lambert- W rate identity were at most 2.23×10^{-16} . These small discrepancies are internal consistency checks, not certified error bars.

The quadrature explicitly separates the compact boundary from the exponential tail. In particular, it does not ask a generic adaptive integrator to find a short exponential boundary layer inside an interval whose length is of order n . The remaining profile-tail mass in the principal dyadic calculations is bounded by e^{-82} before its use in the conditional formulas.

From the package directory, the computations and article can be rebuilt with

```
python code/verify.py
python code/make_figures.py
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex
```

The pre-generated figures and table fragments are included. Rebuilding only the PDF requires no Python execution and no network access. The JSON receipt records the numerical environment, check results, and their scope. Platform and library changes may alter the last displayed digits.

10 Further research questions and topics

The geometric-prefix critical-complement problem is resolved by the three main theorems. The following questions are not presented as solved. Some extend the same explicit repository frontier to settings where the proof mechanisms no longer transfer automatically.

Question 1 (Arbitrary observation masks). For general subsets of geometric coordinates, can the crossover be classified by a complement cumulant function rather than the number m ? A mask may hide isolated large-cap coordinates, a boundary cluster, or both. A variance fraction suffices for the qualitative endpoint, but the present theorem shows that the limiting variance fraction alone does not determine sharp overlap rates in a singular regime. An appropriate replacement for the Gamma rate function should depend on the hidden cap profile.

Question 2 (General summable weights). Which positive summable weight sequences admit an analogue of the signed correction with bounded exponential variation? For polynomial weights, the boundary layer has a growing number of coordinates and need not be an $O(1)$ correction to a Gamma law. Prove a sharp critical-complement theorem in at least one such nongeometric class, and identify whether the logarithmic crossover scale survives.

Question 3 (Second-order crossover terms). In the window $m \sim c \log n$, the first neglected terms in (6.5) contain $\log \mathcal{H}(1 - 1/y)$, integer rounding of m , and the difference between $\log(n/m)$ and $\log n$. Derive the complete next-order overlap expansion, uniformly for c in compact subsets of $(0, \infty)$, and exhibit the reappearance of the geometric phase. The leading theorem gives the correct starting points for both likelihood crossings.

Question 4 (A sharp asymptotic for the compact deficit). Determine the asymptotics of $\Delta_m(a)$ as $a \downarrow 0$ from the small-argument Fabius expansion, including phase oscillations and an explicit error bound. Equation (5.1) reduces the slow fixed-complement correction to a concrete one-dimensional inverse-level problem. This would turn (2.13) into a fully expanded endpoint transseries rather than a formula retaining one exact integral.

Question 5 (A moving geometric ratio). Let $q = q_n \uparrow 1$. The uniform exponential-variation constants in Lemma 4.1 may then diverge and the boundary tail ceases to be a fixed-size perturbation. Find the joint scaling of $1 - q_n$, n , and m that separates Gamma-type, boundary-dominated, and intermediate limits. The fixed- q theorem should not be used as a uniform assertion in this regime.

Question 6 (Other endpoint densities). Replace the uniform coordinates by densities behaving like $c_j x^{\alpha_j - 1}$ near zero. Exponential tilting suggests Gamma coordinates with varying shapes. Under what quantitative regularity hypotheses does an explicit complement rate function yield a version of (2.15)? Even the constant-shape case requires tracking which boundary correction is genuinely bounded and which alters the large-deviation exponent.

Question 7 (Several simultaneous constraints). For two or more weighted sums constrained to a small orthant, the likelihood becomes a multivariate boundary convolution. Determine the analogue of

$I(R_m; R_m + E)$ and classify the overlap by the volume of a multivariate density level set. The geometry of that level set, not just a determinant of a covariance matrix, should matter at critical complement sizes.

Question 8 (Equality, windows, and inequality conditioning). Give a unified theorem when T_n is conditioned on an interval of width w_n immediately below μ_n , allowing $w_n \rightarrow 0$, $w_n \rightarrow w$, and $w_n \rightarrow \infty$. The exponential slack in this paper should be replaced by a truncated kernel or by exact-level conditioning in the appropriate limits. Determine precisely when the leading logarithmic overlap mechanism changes.

Question 9 (Phase entropy and comparison inequalities). The phase derivative of K_m is explicit, but the derivative and possible monotonicity of h_m in ρ have not been settled here. Find useful analytic bounds on $h_m - 1$ and on the covariance in (2.9), uniformly over a phase cell. Such bounds would control information loss without numerical Fourier inversion.

Question 10 (Certified computation and formalization). Construct an interval-certified evaluator for g_m , h_m , Δ_m , and $\mathcal{E}_m$, and formalize the likelihood and signed Gamma identities in Lean. A complete formalization should separate measure-theoretic conditioning from the asymptotic density estimates, and should prove uniform tail bounds rather than importing numerical plots as certificates. Quantitative constants suitable for verified finite- n bounds remain a practical target.

11 Proof status and a formalization roadmap

The article provides conventional proofs of Theorems 2.1, 2.2, and 2.3, including the displayed hypotheses, constants, and remainders. The principal imported analytic fact is convolution preservation of log-concavity. The remaining density estimates are derived from the exact signed convolution and Stirling's formula. We do not invoke an unspecified local limit theorem with an unverified uniformity range.

For integration into the repository, a sensible dependency order is: first the countable product tilt and exact likelihood; then the truncated-exponential/signed-Gamma identity and its weighted variation bounds; then central density estimates and the complement saddle estimate; finally the overlap and entropy limits. The finite algebra and the countable measure-theoretic interface are logically independent of the analytic large- n step. This separation should make auditing and alternative proofs easier.

No new Lean or Rocq proof file is included, and no repository-wide build or axiom audit was run. The included Python programs test numerical consequences; they do not certify the theorems. The prior research manuscripts are themselves distinguished from the repository's formally proved Fabius development. None of those statuses is upgraded merely because this paper builds on the same topic.

The resolved scope is exact: fixed geometric ratio, a fixed tilt phase, inequality conditioning at the tilted mean, and an observed prefix whose missing bulk size is sublinear. Within that scope the qualitative singular endpoint has been replaced by explicit laws. Beyond it, Section 10 records the remaining problems rather than silently extrapolating the results.

A Numerical identities used by the implementation

A.1 Finite signed inclusion–exclusion

For a finite collection of caps $a_1, \dots, a_k$, the density of the sum of independent p_{a_j} variables is exactly

$$\frac{1}{\prod_{j=1}^k (1 - e^{-a_j})} \sum_{J \subseteq \{1, \dots, k\}} (-1)^{|J|} e^{-a_J} \gamma_k(s - a_J), \quad a_J = \sum_{j \in J} a_j. \quad (\text{A.1})$$

This follows by expanding the product of signed measures in (4.3). The implementation retains only the caps below a chosen numerical threshold. The remaining variables are full exponentials, but still contribute to the Gamma shape k . Thus the number of signed terms depends on the number of moderate caps, not on the effective dimension n .

A variable with law p_a is an exponential conditioned to be at most a . It can be coupled with an unconditioned exponential with mismatch probability e^{-a} . A union bound gives the recorded omission bound $\sum e^{-a_j}$ for the replaced coordinates. Formula (A.1) is exact before this computational simplification.

A.2 Fourier inversion of the boundary CDF

Let $V = \sum_{\ell \geq 0} c q^\ell U_\ell$ have support $[0, A]$, $A = c/(1 - q)$. Its Fourier transform under the convention $\widehat{f}(\omega) = \mathbb{E} e^{-i\omega V}$ is

$$\widehat{f}(\omega) = e^{-i\omega A/2} \prod_{\ell \geq 0} \operatorname{sinc}\left(\frac{\omega c q^\ell}{2}\right), \quad \operatorname{sinc}(z) = \frac{\sin z}{z}. \quad (\text{A.2})$$

The removable value at zero is one. The code uses a periodic interval of length $2A$, which contains the support without an interior wrap, and integrates its Fourier series by dividing each nonzero coefficient by $i\omega$. The zero-frequency term contributes $u/(2A)$. Subtracting the primitive's value at zero produces the CDF. The density g_m is then recovered by multiplying the CDF by $e^{K_m - u}$.

Finite Fourier truncation, truncation of the infinite product, and interpolation mean that the implemented CDF is not an exact real function. Increasing the grid supplies a convergence check; it is not an interval proof. The tail identities used after $u = A_m$ are analytic, so no Fourier evaluation is needed there. In particular,

$$\begin{aligned} \int_{A_m}^{\infty} g_m(u) \, du &= e^{K_m - A_m}, \\ - \int_{A_m}^{\infty} g_m(u) \log g_m(u) \, du &= e^{K_m - A_m} (A_m + 1 - K_m). \end{aligned}$$

These formulas improve both stability and reproducibility.

B Elementary checks on the crossover roots

The derivative $I'(y) = 1 - 1/y$ is negative on $(0, 1)$ and positive on $(1, \infty)$. Since $I(1) = 0$ and I tends to infinity at both ends, there are exactly two roots for every positive level. The equation $I(y) = z$ is equivalent to $(-y)e^{-y} = -e^{-1-z}$, proving (2.14).

For $z \downarrow 0$, write $y = 1 + v$ and expand $I(1 + v) = v^2/2 - v^3/3 + O(v^4)$. Thus

$$y_{\pm}(z) = 1 \pm \sqrt{2z} + O(z), \quad y_+(z) - y_-(z) \sim 2\sqrt{2z}.$$

For $z \rightarrow \infty$, the lower root tends to zero and satisfies $y_- = e^{-1-z+y_-} \sim e^{-1-z}$. The upper root satisfies $y_+ = z + 1 + \log y_+$, hence $y_+/z \rightarrow 1$. These observations justify all endpoint limits used in Section 6.

They also show that the leading equivalent remains positive and tends to zero for every $m = o(n)$: on the Gaussian side it is controlled by $\sqrt{(m/n) \log(n/m)}$, and on the other two sides it is of order $\log n / \sqrt{n}$. The crossover formula is therefore compatible with the predecessor's qualitative conclusion $d_{\text{TV}} \rightarrow 1$, while supplying information that that conclusion alone does not contain.

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